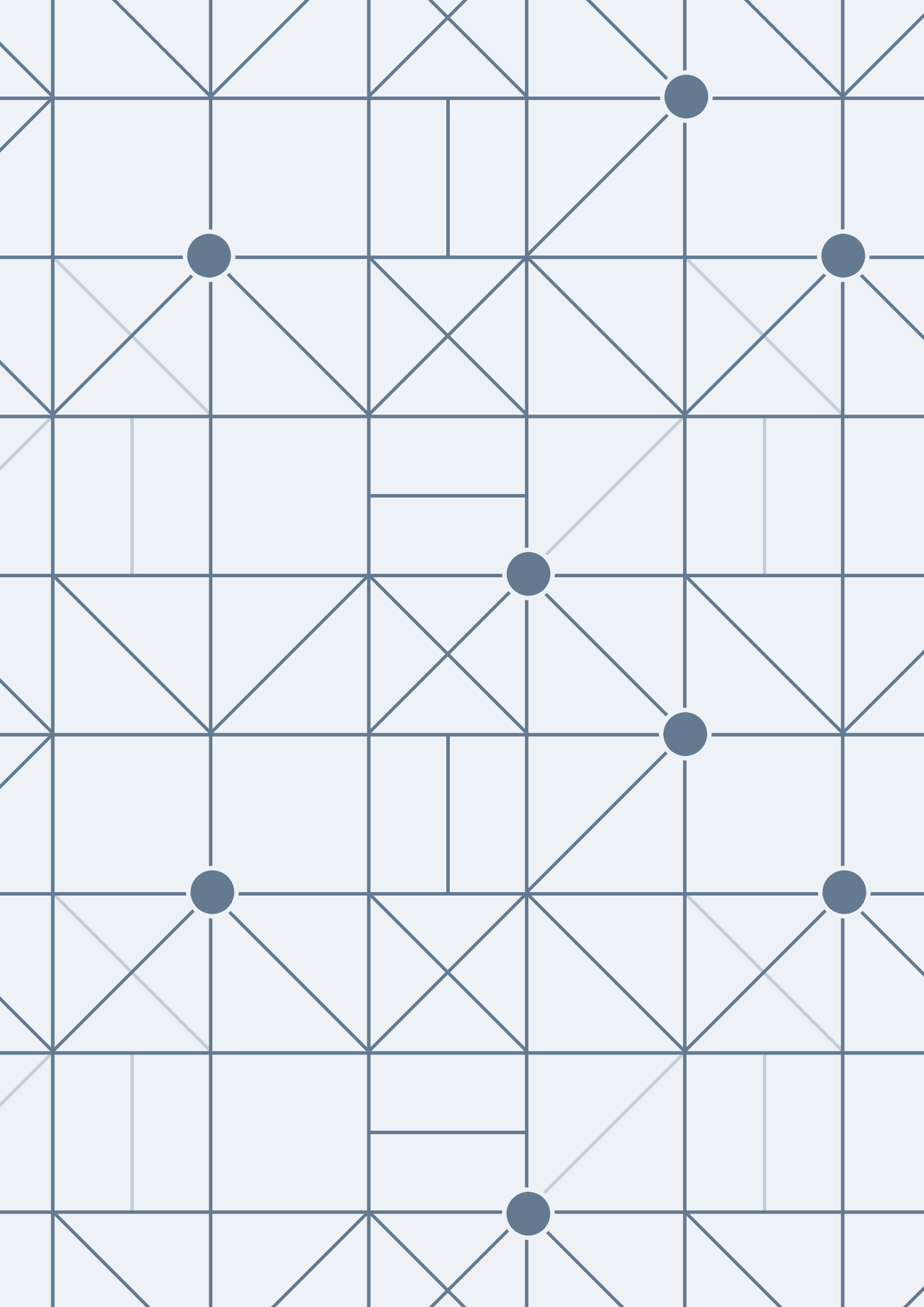




Liouville's Theorem

The density of an ensemble of systems of thermodynamic equilibrium remains constant as they evolve according to Hamilton's equations of motion.

$$\frac{\partial \rho}{\partial t} + \{\rho, H\} = 0 \Rightarrow \frac{d\rho}{dt} = 0$$

$$\{\rho, H\} = 0$$

$\frac{1}{2\pi} \int_{-\pi}^{\pi} \delta(x + \frac{1}{2}\pi) dx$

Integrando por partes

$\int_{-\pi}^{\pi} \delta(x + \frac{1}{2}\pi) dx = 1$

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Liouville's Theorem

$\{\rho, H\} = 0$

Particular solution $\Rightarrow \rho = \text{const}$

Microcanonical ensemble

Other ensembles

Canonical ensemble

Grand canonical ensemble

$\rho \sim e^{-E/KT}$

$\rho \sim e^{-E/KT + \mu N/KT}$

Application example

A free particle of mass m is constrained to move in one dimension, with a Hamiltonian given by $H = p^2/2m$. Let's verify that the area of the rectangular enclosure shown in Fig. 6.6 remains invariant during the natural evolution of the system.

Figure 3.6: Region of the phase space where a free particle of mass m is constrained to move.

$\alpha(q_0, p_0) \rightarrow \alpha(q_0 + \Delta q, p_0)$

$\alpha(q_0, p_0) \rightarrow \alpha(q_0, p_0 + \Delta p)$

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Quantum Formulation of the First Postulate

First Postulate - classical formulation

"The values of macroscopic physical quantities at time t are given as the averages of microscopic dynamic functions."

Let's remember that the **expected value** of an operator $\hat{B}$ when a quantum system is in the pure state $|\psi\rangle$ is given by:

$$\bar{B} = \langle \psi | \hat{B} | \psi \rangle$$

It is a quantum average. It is the mean result we would obtain if we performed many measurements of $\hat{B}$ after preparing our system in the quantum state $|\psi\rangle$.

The quantum average is NOT the average in the ensemble.

- The expected value obtained in Quantum Mechanics represents a **first average**. It is an average that depends on the quantum state $|\psi\rangle$ in which the system is located: **pure state**.
- However, the only information we have about the system is its macrostate. We know nothing about the specific microstate it is occupying: **mixed state**.

$|\psi\rangle = \sum_k c_k |\phi_k\rangle$

$\langle \psi | \hat{B} | \psi \rangle = \sum_k \sum_l c_k^* c_l \langle \phi_k | \hat{B} | \phi_l \rangle$

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Density Matrix

To obtain the matrix expression of the operator, it is necessary to write all the functions $\psi^{(n)}(r)$ in the orthonormal basis $\{|\phi_n\rangle\}$ of the Hilbert space:

$$|\psi^{(n)}(r)\rangle = \sum_m c_m^{(n)} |\phi_m\rangle$$

$$\langle \psi^{(n)}(r) | = \sum_m c_m^{(n)*} \langle \phi_m |$$

Some steps...

$$\hat{B}^{(n)}(r) = \sum_m c_m^{(n)*} \langle \phi_m | \hat{B} | \phi_m \rangle c_m^{(n)}$$

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$$\sum_{i=1}^n \left(\sum_{j=1}^n a_{ij} b_{ij} \right) = \sum_{i=1}^n (a_{i1} b_{i1} + a_{i2} b_{i2} + \dots + a_{in} b_{in}) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ij}$$

$$\text{Tr} \left[\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \right] = \text{Tr} \left[\begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix} \right] = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ji}$$

$$\langle \hat{b} \rangle = \text{Tr} [\hat{\rho} \hat{b}]$$

density matrix, set of probabilities

Only the diagonal of $\hat{b}$ contribute to the result, because the diagonal components of diagonal and non-diagonal of $\hat{\rho}$ cancel if

by definition

$$p_{ii} = \frac{1}{2\pi} \int_0^{2\pi} e^{i\theta} e^{-i\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} 1 d\theta = 1$$

$$\lim_{n \rightarrow \infty} p_{ii} = \frac{1}{2\pi} \int_0^{2\pi} e^{i\theta} e^{-i\theta} d\theta = \frac{1}{2\pi} \int_0^{2\pi} 1 d\theta = 1$$

$$\lim_{n \rightarrow \infty} p_{ij} = \frac{1}{2\pi} \int_0^{2\pi} e^{i\theta} e^{-i\theta} d\theta = 0$$

$$\hat{b} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\text{Tr} \left(\hat{\rho} \hat{b} \right) = \frac{1}{2\pi} \int_0^{2\pi} \left(\begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \right) d\theta = \frac{1}{2\pi} \int_0^{2\pi} \begin{pmatrix} a \cos \theta + b \sin \theta & a \sin \theta + b \cos \theta \\ -c \sin \theta + d \cos \theta & -c \cos \theta + d \sin \theta \end{pmatrix} d\theta$$